

CHANDRASHEKAR BONTHU

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GitHub | Portfolio | LinkedIn

PROFESSIONAL SUMMARY

Backend-focused Computer Science graduate with hands-on experience designing, developing, testing, and deploying production-oriented applications using Node.js, TypeScript, Express.js, PostgreSQL, MongoDB, Redis, and Python. Skilled in REST APIs, JWT authentication, RBAC, MFA, API validation, transactional workflows, event-driven microservices, real-time communication, background jobs, Docker, CI/CD, observability, and OpenAPI. Built end-to-end systems spanning telemedicine, distributed services, real-time project management, e-commerce, and invoicing. Solved 200+ Data Structures and Algorithms problems with strong foundations in OOP, DBMS, Operating Systems, and Computer Networks.

TECHNICAL SKILLS

Languages: JavaScript, TypeScript, Python, SQL

Backend: Node.js, Express.js, FastAPI, REST APIs, JWT, RBAC, MFA, WebSockets, Middleware, API Validation

Databases: PostgreSQL, MongoDB, MongoDB Atlas, MySQL, Prisma, Mongoose

Distributed Systems: Microservices, API Gateway, NATS JetStream, Event-Driven Architecture, Outbox Pattern, Idempotency

Caching & Jobs: Redis, BullMQ, node-cron

DevOps & Tools: Docker, Docker Compose, AWS, Render, Vercel, GitHub Actions, Git, GitHub, Postman, Swagger/OpenAPI

Observability: Prometheus, OpenTelemetry, Jaeger

PROJECTS

Amrutam Telemedicine Backend | Node.js, TypeScript, PostgreSQL, Prisma, Redis

- Built a production-oriented telemedicine API covering authentication, RBAC, TOTP MFA, doctor availability, consultation booking, prescriptions, audit trails, and admin analytics.
- Implemented concurrency-safe booking and idempotent writes with Redis/BullMQ background jobs; added Prometheus metrics, OpenTelemetry/Jaeger tracing, automated tests, Docker Compose, GitHub Actions CI, and OpenAPI documentation.

GitHub

Microservices User & Notification System | Node.js, Express.js, MongoDB, NATS

- Designed an API Gateway, User Service, and Notification Service with Node.js, Express.js, MongoDB, JWT, and Docker Compose using asynchronous NATS JetStream communication.
- Implemented the Outbox pattern, durable consumers, acknowledgements, retries, idempotent event processing, Zod validation, rate limiting, Helmet, CORS, health checks, and internal service authentication.

GitHub

Real-Time Client Project Dashboard | React, TypeScript, Node.js, PostgreSQL

- Developed role-based Admin, Project Manager, and Developer workflows using React, Node.js, Express.js, PostgreSQL, Prisma, and Socket.IO with server-side ownership and assignment authorization.
- Implemented rotating JWT refresh tokens with HttpOnly cookies, real-time notifications, missed-event recovery, Zod validation, security middleware, rate limiting, database indexes, and scheduled overdue-task processing.

GitHub | Live Demo

E-Commerce Backend API | Node.js, Express.js, MongoDB

- Developed 15+ REST APIs for authentication, products, carts, orders, stock validation, and admin workflows using Node.js, Express.js, MongoDB, and Mongoose.
- Implemented JWT/RBAC, bcrypt, Joi validation, pagination, filtering, centralized error handling, Swagger documentation, Postman testing, MongoDB Atlas integration, and Render deployment.

GitHub | Live API | Swagger

Modern Invoicing System | React, Node.js, Express.js, MongoDB

- Built a full-stack invoicing application with JWT authentication, user-scoped client/invoice CRUD, status workflows, filtering, date-range queries, and automatic financial calculations.
- Implemented authorization and validation with MongoDB/Mongoose, image upload and processing using Multer/Sharp, and deployed the React frontend on Vercel and backend on Render.

GitHub | Live Demo

EDUCATION

B.Tech in Computer Science Engineering

Malla Reddy College of Engineering and Technology, Hyderabad

CGPA: 7.65

2022 – 2026

CERTIFICATIONS

Deloitte Cyber Job Simulation | Deloitte Data Analytics Job Simulation | Quantum Software Engineering Job Simulation